Inverted, Re-Ranked and Unstable: What Natural Zero-Day Traffic Reveals About Learned Network Intrusion Detectors

Hafsa [surname]*, [co-authors]
*[Department, University, City, Country]
Email: [email]

***Abstract*—Operators deploy learned network intrusion detection systems (NIDS) to catch attacks absent from their training data, yet zero-day performance is usually measured once, by class hold-out, with novelty scores borrowed from multi-class image recognition. We present a pre-registered, leakage-controlled evaluation study, not a new detector: 18 detectors from five families on two recurrent backbones, five seeds each, on four NetFlow-v3 datasets and LYCOS-IDS2017. Besides leave-one-attack-class-out (LOACO) we use natural temporal novelty, where attack classes first appear after training, at rolling cut-points that yield 45 zero-day units over 19 attack classes. Four findings hold on both backbones. Max-softmax and energy scores are misaligned with novelty in binary detectors: a confidently detected attack reads as normal (MSP 0.163 vs attack probability 0.999 on the same models). Family rankings change with protocol, dataset and cut-point: pooled over rolling cuts, benign-only detectors lead and XGBoost, which matches the best novelty family in four of seven fixed evaluations, trails them. Natural novelty raises the seed variability of representation-distance scores 2.7–17.7-fold. Pooled AUCs of 0.53–0.75 mean at most 18% detection at 1% false-positive rate. We report four failed hypotheses and release all splits and code.**

***Index Terms*—security management, service assurance, machine learning, NetFlow monitoring, intrusion detection, zero-day attacks**

I. Introduction

Anetwork operator who deploys a learned network intrusion detection system (NIDS) is buying protection against attacks that are not in its training data. Before trusting the detector, the operator needs evidence that it raises an alarm when a new attack class first appears, at a false-alarm rate the security team can absorb, and that this behaviour survives the routine retraining of an operational service. A detector that misses novel attacks creates false assurance; one whose zero-day coverage changes from one retraining to the next cannot be operated predictably. The management task is therefore twofold: acceptance testing of a detection service before deployment, and re-validation after every retraining.

The problem. The evidence operators receive is usually a single training run evaluated by leave-one-attack-class-out (LOACO): one attack class is removed from a randomly split benchmark and the detector must flag it at test time [22], [23], [20]. Novelty is often scored with confidence scores designed for multi-class image recognition, such as maximum softmax probability (MSP) [37] and energy [38], and summarised by ROC AUC. In deployment, however, a new class arrives later in time, together with drifting benign traffic; the detector is retrained repeatedly; and only the detection rate at a low false-alarm rate matters. Security-ML research has shown that evaluation shortcuts inflate results [1], [2], [3], [4], and recent work has begun to study time-aware NIDS splits [25], [15], [16].

The gaps. Four questions that matter to an operator remain open. (G-1) *Score validity:* MSP-type scores have been observed to perform poorly on unseen attacks in multi-class NIDS [13], [14], but no work explains why, or whether they are valid at all for the common two-class benign-versus-attack detector. (G-2) *Rankings:* studies apply class hold-out and temporal splits to a single proposed method [18], [19] or rank supervised classifiers only [15]; whether the ranking of detector *families* survives the move from LOACO to natural novelty is unknown, and prior comparisons disagree on whether supervised models or novelty detectors handle unknown attacks better [21], [17]. (G-3) *Stability:* prior work reports at most aggregate run-to-run variance [14]; whether a zero-day result is stable across retraining, per attack class, is unknown. (G-4) *Operating point:* zero-day results are reported as AUC, which does not show what fraction of a new attack is caught at an operational false-positive rate.

Research questions. **RQ1** Which novelty scores are valid for binary NIDS, and why do confidence scores fail? **RQ2** Do detector-family rankings agree across protocols, datasets and cut-points, and do dedicated novelty detectors beat a supervised classifier? **RQ3** How stable are zero-day results across training seeds? **RQ4** What do zero-day AUCs mean at an operational false-positive rate?

Approach. This is an evaluation study: we propose no new detector. We build a pre-registered, leakage-controlled benchmark on five public datasets, four NetFlow-v3 datasets [31] and LYCOS-IDS2017 [8]. On day-scheduled captures, where attack classes occupy separate days [16], a time-ordered split leaves classes that appear only after the training period; we use them as a natural zero-day protocol, at a primary split and at four rolling cut-points, alongside LOACO. We evaluate 18 detectors in five families, built on two recurrent backbones, over five seeds, from one code commit.

Contributions.

- **C1 Confidence scores are misaligned with novelty (RQ1).** We show why MSP and energy fail in binary

NIDS: both are functions of the logit margin's magnitude, so a confidently detected novel attack reads as in-distribution, and no temperature calibration can change their ranking. On both backbones MSP scores 0.16 on LOACO UNSW-NB15 where the attack probability scores 0.999 on the same models.

- **C2 Rankings depend on protocol, dataset and cut-point (RQ2).** Both backbones give the same leading family in each of seven evaluations, but the leader changes between evaluations and across rolling cut-points. Pooled over 45 rolling units, benign-only detectors lead (0.748 AUC); XGBoost, which matches or beats the best novelty family in four of seven fixed evaluations, trails them by -0.076.
- **C3 Hidden instability (RQ3).** Natural novelty raises the per-class seed standard deviation of representation-distance scores 6.1–17.7$\times$ (first backbone) and 2.7–2.8$\times$ (second), concentrated on ToN-IoT; seeded benign-only detectors stay below 0.02.
- **C4 AUC overstates zero-day coverage (RQ4).** At 1% false-positive rate, no family detects more than 18% of rolling-cut novel windows, although family AUCs reach 0.75.
- **C5 Pre-registration and release.** To our knowledge this is the first pre-registered NIDS benchmark; we report its four failed hypotheses, ablations, a comparison with previously reported results, and release splits, code and amendments.

Sections II–III give background and related work, Sections IV–V the benchmark, Section VI the results and Section VII the discussion with an assurance checklist.

II. Background

Flow-based NIDS and windowing. Flow-based NIDS summarise each connection as a record of counters and timings. Sarhan et al. converted public captures into a common NetFlow feature set [32]; version 3 adds per-flow start and end timestamps [31], which make strictly time-ordered splits possible. Sequence detectors consume windows of T consecutive flows. Overlapping windows can place near-identical samples in training and test, and flows of one attack episode can straddle the split; grouping flows into time blocks before windowing and deduplicating windows remove both risks.

Zero-day protocols. LOACO removes one attack class from the training pool and tests on benign windows plus that class, under random splits (here grouped by 1-hour blocks). *Natural temporal novelty* splits flows in time: attack classes that occur only after the training period are unseen by construction, and benign traffic drifts as it would in operation. *Purged block LOACO*, for short captures, splits short time blocks within each day and drops blocks adjacent to test blocks [45]. LOACO asks whether a detector generalises to a missing class under the training distribution; natural novelty asks the operator's question, under real temporal shift.

Novelty scores and the binary pitfall. Classifier-confidence scores read the output layer: MSP scores $1 - \max_k p_k$ [37] and energy $-\log \sum_k e^{f_k}$ over logits f_k [38]. *Representation-distance* scores measure how far hidden features lie from benign training features [39], [40]; *benign-only* detectors never see attacks [41], [42]; *contrastive* detectors such as CLAD [22] learn an embedding from benign and known attacks. In a two-class detector with logits f_0 (benign) and f_1 (attack), let $\Delta = f_1 - f_0$ and $p = \sigma(\Delta)$. Then

$$s_{\mathrm{MSP}} = 1 - \max(p, 1-p) = \sigma(-|\Delta|), \qquad s_{\mathrm{attack}} = \sigma(\Delta). \tag{1}$$

s_{MSP} is a decreasing function of $|\Delta|$ alone: it rewards uncertainty, not the predicted class. A novel attack confidently classified as *attack* ($\Delta \gg 0$) receives the lowest possible novelty score, exactly like a confidently recognised benign window. Temperature scaling, $\Delta \mapsto \Delta/\tau$, preserves the order of $|\Delta|$, so calibration cannot change MSP's AUC; energy, $-\log(e^{f_0} + e^{f_1})$, is likewise symmetric in the two classes. The attack probability s_{attack} is monotone in Δ and keeps the class. In a binary NIDS, successful detection of a novel attack therefore turns into a missed novelty under MSP and energy.

III. Related Work

Evaluation pitfalls and dataset quality. Sommer and Paxson argued that anomaly detection outside a closed world is harder than benchmarks suggest [1]; Arp et al. catalogued ten pitfalls in security ML [2]; TESSERACT showed that ignoring time inflates malware-detection results [3]. For NIDS, Apruzzese et al. found that evaluations rarely reflect operational conditions [4], and Husák et al. that research favours ML metrics over operational outcomes [5]. Dataset audits found labelling and flow-construction errors in CIC-IDS2017/2018 [6], [7], corrected in LYCOS-IDS2017 [8], design smells that bias experiments [9], and metadata features that let models shortcut the task [10].

Zero-day and open-set NIDS. A TNSM survey concludes that ML detectors handle unknown attacks poorly [11]. Sarhan et al. framed zero-day detection as zero-shot learning [20]; Zoppi et al. compared 47 algorithms on unknown attacks and found unsupervised meta-learners strongest [21]; Wilkie et al. proposed CLAD [22]; Jodayree et al. combined a transformer with energy scoring [23]; Thi and Gurusamy detect open-set attacks from benign traffic only [24]; and Baye et al. benchmarked deep open-set recognition at NOMS [12]. Softmax baselines scored below chance on held-out attacks in multi-class NIDS [13], and softmax entropy and energy trailed uncertainty-aware models [14], without an explanation tied to the binary setting. All of these works evaluate under class hold-out or in-distribution.

Drift and temporal evaluation. INSOMNIA showed that accounting for drift is vital in NIDS [25]; ReCDA [26] and continual-learning NIDS [27] adapt to drift. Abdessamad and Tazi found that temporal splits on ToN-IoT leave unseen classes in test and change classifier rankings [15]; Moczkodan and Ragab found a chronological CIC-IDS2017 split degenerate because attacks were scheduled by day [16]; Mesca et al. compared classifiers under several protocols on industrial

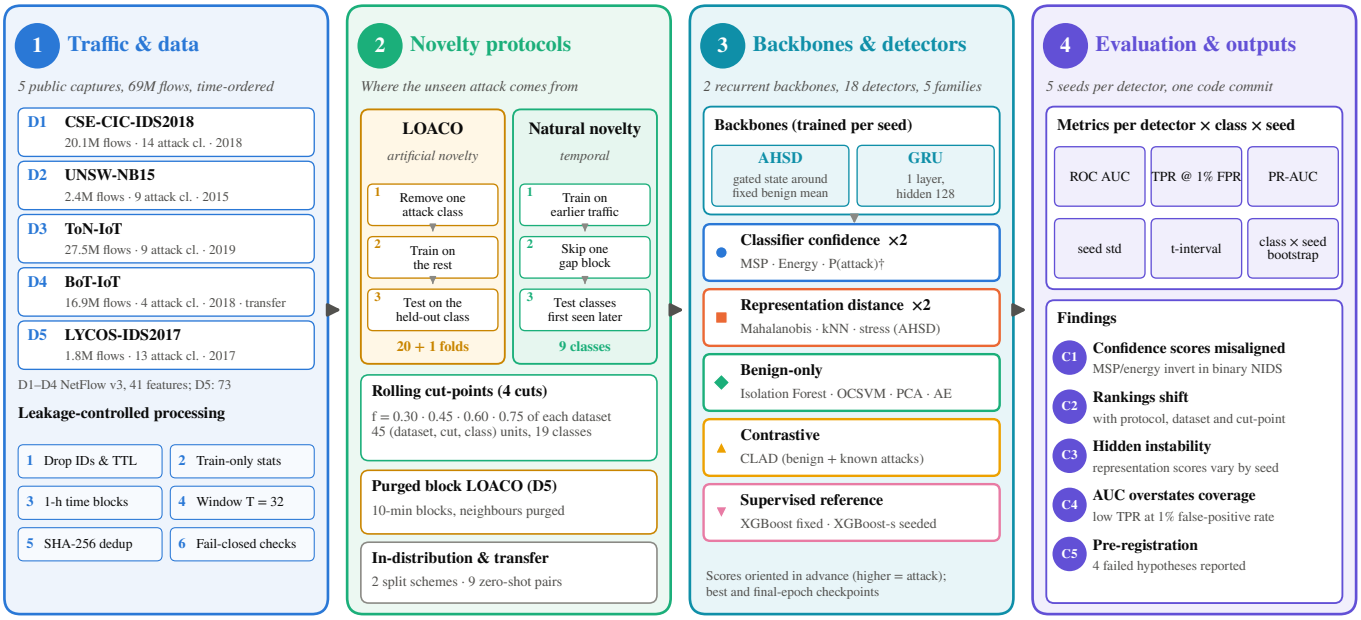


Fig. 1. Benchmark methodology. (1) Five public datasets enter with checksummed manifests. (2) Seven processing steps turn flows into deduplicated, time-blocked windows and fail-closed checked splits. (3) Two recurrent backbones supply classifier-confidence and representation-distance scores; benign-only, contrastive and XGBoost detectors are trained on the same splits. (4) Five evaluation settings, each over five seeds, produce findings C1–C5.

TABLE I
POSITIONING AGAINST REPRESENTATIVE PRIOR WORK (–: NOT ASSESSED).

Work	Zero-day protocol	X-data	Families	Seed var.	Op. point	Pre-reg.
Sarhan et al. [20]	zero-shot hold-out	no	supervised	–	DR	no
Zoppi et al. [21]	unknown attacks	no	sup./unsup./meta	–	recall	no
Wilkie et al. [22]	LOACO	no	contrastive, anomaly	–	–	no
Nayak [18]	held-out + chrono.	no	one model	5 seeds (agg.)	TPR@1%	no
Abdessamad & Tazi [15]	temporal	no	supervised	–	–	no
This work	**LOACO + natural + rolling**	**9 pairs**	**5, 2 backbones**	**per class**	**TPR@1%**	**yes**

TABLE II
DATASETS. WINDOWS ARE COUNTED AFTER GLOBAL DEDUPLICATION.

ID	Dataset	Flows	Windows	Att. cl.	Role
D1	NF-CSE-CIC-IDS2018-v3	20,115,529	1,255,180	14	LOACO, NN, transfer
D2	NF-UNSW-NB15-v3	2,365,424	147,801	9	LOACO, transfer
D3	NF-ToN-IoT-v3	27,520,260	1,553,191	9	LOACO, NN, transfer
D4	NF-BoT-IoT-v3	16,933,808	890,852	4	transfer target
D5	LYCOS-IDS2017	1,837,498	114,783	13	NN, purged LOACO

traffic [17]; and ThreatFormer-IDS [18] and Alhalangy and Eltayeb [19] evaluate one proposed detector under chronological and held-out-family splits. Cross-dataset studies report near-chance transfer [29], [30]. None compares detector families under class hold-out versus natural novelty, across seeds and at an operating point (Table I).

IV. Benchmark Methodology

Fig. 1 shows the benchmark in four stages. Data, splits, backbones and detectors were fixed in advance, and every change was recorded as a dated, hashed amendment before the runs it governs (Section IV-E).

A. Stage 1: Input datasets

All datasets are public, were downloaded from their official sources and are recorded with SHA-256, row count and column list (Table II). D1–D4 are the NetFlow-v3 releases [31], which re-extract each capture's packet traces into 53 NetFlow features with millisecond timestamps [32]. **D1 CSE-CIC-IDS2018** [33]: cloud-hosted enterprise testbed with scripted users and attackers, 14 Feb–2 Mar 2018; 14 attack labels in six families; known labelling errors [7]. **D2 UNSW-NB15** [34]: cyber-range traffic generator mixing normal activity with nine synthetic attack families, 22 Jan–18 Feb 2015. **D3 ToN-IoT** [35]: IoT/IIoT testbed with edge, fog and cloud layers and nine attack types, 23–29 Apr 2019. **D4 BoT-IoT** [36]: simulated IoT botnet attacks, May–Jun 2018; only 1,315 benign windows, so a transfer target only. **D5 LYCOS-IDS2017** [8]: the CIC-IDS2017 traces (3–7 Jul 2017) re-processed with a corrected flow extractor and labels [6]; we ran the authors' script unmodified and keep its 73 features.

B. Stage 2: Leakage-controlled processing

1) *Audit and drop.* Record rows, types, missing values and class counts; drop identifiers (addresses, ports, DNS query ID) [10] and minimum/maximum TTL, a label-leakage artifact; keep timestamps only for ordering.
2) *Clean with training statistics only.* Infinite values become missing and take the training median; heavy-tailed counts are $\log(1+x)$-transformed; features are standardised; zero-variance features and one of each pair with $|r| \geq 0.999999$ are removed, leaving 41 common features for D1–D4. Nothing is fitted on test or transfer-target data.

3) *Sort and group in time.* Flows are sorted by start time and grouped into 1-hour blocks (10-minute blocks for D5's purged split).
4) *Window and label.* Windows of T=32 flows with stride 16 never cross a block boundary. A window is an attack if any flow is, and its class is the majority attack class.
5) *Deduplicate globally.* Windows are hashed and exact duplicates removed across the whole dataset (9.7% of D3, 15.8% of D4).
6) *Split and balance.* Blocks, never flows or windows, are assigned to splits. Training sets are balanced 1:1 (7,000 + 7,000 windows); validation and test sets are 5:1 benign to attack, capped at 2,500 benign and 500 attack windows per class.
7) *Fail-closed checks.* The pipeline stops on any failed check: row and block disjointness, deduplication, balance, class composition, feature lists. 538 checks ran at preprocessing for D1–D4, and 12–32 for every derived split; all passed.

C. Stage 3: Backbones and detectors

Backbone A (AHSD). Each flow is embedded as $z_t = \mathrm{LayerNorm}(Wx_t) \in \mathbb{R}^{128}$. A state h_t, initialised at a fixed benign reference E (the mean benign training embedding), follows $h_t = h_{t-1} + \alpha_t \odot (z_t - E) - \beta_t \odot (h_{t-1} - E)$ with gates $[\alpha_t, \beta_t] = \sigma(W_g[z_t, h_{t-1}])$; *stress* is $\frac{1}{TD}\sum_{t,d} |h_{t,d} - E_d|$. A two-layer MLP head reads $[h_T, \bar{h}, \text{stress}]$. *Backbone B (GRU).* The same embedding feeds a one-layer GRU (hidden 128); the same head reads $[h_T, \bar{h}]$. Both are used as backbones only; neither is proposed as a detector.

Detectors. Eighteen detectors score each test window; higher always means more anomalous, fixed before any run.

- *Classifier-confidence* (per backbone): MSP, energy, and P(attack), the softmax attack probability, added after the last confirmatory gate and marked †.
- *Representation-distance* (per backbone): Mahalanobis distance to a Ledoit–Wolf Gaussian on benign penultimate features [39]; distance to the 10th benign nearest neighbour of ℓ_2-normalised features [40]; and, for AHSD, stress.
- *Benign-only*: Isolation Forest (200 trees) [41], one-class SVM (RBF, ν=0.1) [42], PCA reconstruction (95% variance) and an autoencoder.
- *Contrastive*: CLAD [22], implemented from its paper.
- *Supervised reference*: XGBoost [43] on flattened windows (300 trees, depth 6, learning rate 0.1), deterministic; and XGBoost-s, the same with row and column subsampling 0.8 and the run's seed.

Families are computed per backbone and never pooled across backbones.

D. Stage 4: Evaluation protocols

grouped_random shuffles blocks into 60/20/20 train/validation/test; *temporal_gap* takes the first 60% of blocks in time for training, discards one gap block, and uses the next 20% for validation and the last 20% for test.

- *LOACO* (grouped_random): a class is held out if it has ≥200 training-pool windows; training and validation exclude every window containing a flow of that class.
- *Natural temporal novelty* (temporal_gap): classes that are the majority class of ≥20 test windows and of no training window are scored against test benign windows. Validation is the last 20% of the training period (train-tail validation).
- *Rolling cut-points*: the natural-novelty rule is re-applied with training ending at a fraction $f \in \{0.30, 0.45, 0.60, 0.75\}$ of each dataset's windows, one gap block, and testing on everything after; the cleaner is refitted on each cut's training rows. Each (dataset, cut, class) is a unit.
- *Purged block LOACO* (D5): 10-minute blocks are split within each day, and blocks adjacent to test or validation blocks are removed [45].
- *In-distribution and zero-shot transfer*: binary detection on both splits; models trained on D1–D3 are applied to every other NetFlow-v3 dataset with all references from the source network.

E. Pre-registration and provenance

Research questions, metrics and gates were written to a file and hashed before each experiment; every result records that hash, the code commit and the dataset hashes. Seven dated, hashed amendments changed the plan before the runs they govern: two because data counts made a frozen rule infeasible, three that set a new direction after a gate returned STOP, one that added P(attack) and the family split, and one that added the second backbone, XGBoost-s, rolling cut-points and operating-point metrics after external review. Four confirmatory gates were evaluated exactly as frozen. The final benchmark is descriptive: it reports intervals, not significance tests.

V. Experimental Setup

Training. Both backbones use AdamW (learning rate 10^{-3}, weight decay 10^{-4}), cosine annealing, gradient clipping at 1.0, batch 256 and 15 epochs, with no per-detector tuning. The checkpoint with the best validation macro-F1 is primary (CLAD: best validation AUROC); the final-epoch state of the same run is scored as a sensitivity analysis.

Evaluations. LOACO covers D1 families, D2, D3 and purged D5 (6, 8, 6 and 1 folds); natural novelty covers D1, D3 and D5 (2, 5 and 2 classes); rolling cut-points add 45 units over 19 unique classes (Table III); in-distribution detection covers four datasets under both splits, and transfer nine source–target pairs.

Metrics and statistics. Seeds are 17, 23, 42, 101 and 202; a seeding regression test runs before every experiment. Per detector and class we report ROC AUC (mean over seeds, Student-t 95% interval with t=2.776), the seed standard deviation, TPR at 1% and 0.1% false-positive rate (FPR) and PR-AUC on the 5:1 test sets. The FPR threshold is set on each test set's benign windows, so TPR@FPR is descriptive, not a

TABLE III
PROTOCOL FEASIBILITY: FEASIBLE LOACO FOLDS PER SPLIT, NATURAL-NOVELTY (NN) CLASSES OF THE PRIMARY TEMPORAL SPLIT, AND ROLLING-CUT UNITS (FEASIBLE CUTS OF 4).

Dataset	LOACO grouped	LOACO temporal	NN	Rolling
D1	6 (fam.)	0	2	15 (4)
D2	8	8	0	0 (0)
D3	6	0	5	15 (3)
D5	0	0	2	15 (3)
D5 purged block	1 (dos_hulk)		–	–

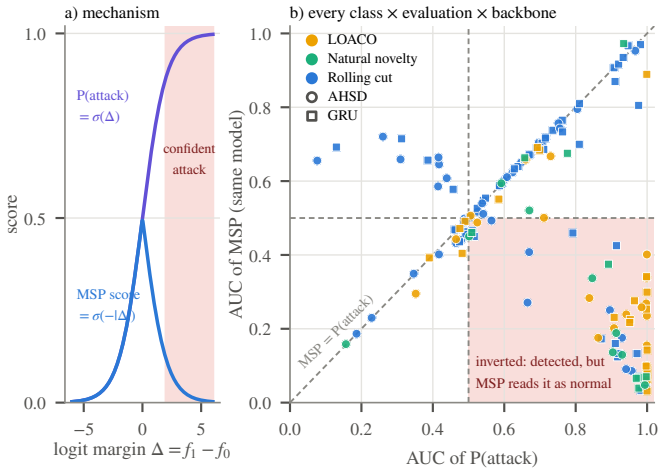


Fig. 2. Why confidence scores invert. (a) In a binary detector the MSP score is $\sigma(-|\Delta|)$: a confidently detected attack (shaded) gets the lowest novelty score. (b) MSP vs P(attack) AUC on the same models for every class, evaluation and backbone; points in the shaded quadrant are attacks the classifier detects but MSP reads as normal.

deployment calibration; 0.1% of 2,500 benign windows is 2.5 windows, so we report it only in the release. Family means average a family's detectors and an evaluation's classes, with a 95% bootstrap interval resampling classes (or rolling units) and seeds independently (2,000 resamples).

Compute and provenance. The 210 final runs used one code commit on a CPU-only cloud container (4 vCPUs, Intel Xeon at 2.10 GHz, no GPU) and took 10.78 h (185 s per run). Report scripts refuse to run if code commits differ.

VI. RESULTS

Values are mean AUC over five seeds unless stated; brackets give 95% intervals.

Evaluation setting. As observed for other day-scheduled captures [16], [15], a time-ordered split places every attack class of D1 and D3 wholly before or wholly after the cut, so temporal LOACO has no feasible fold on D1 and D3 (Table III); natural novelty is then the only zero-day protocol. D2 has no test-only class at any cut; D3 has no benign validation windows at f=0.30. Rolling cuts multiply the natural-novelty evidence from 9 classes to 45 units over 19 classes, 15 per dataset.

A. C1: Confidence scores are misaligned with novelty

The mechanism of Section II appears on both backbones (Fig. 2): almost every point lies on or below the diagonal, and the shaded quadrant collects the attacks that the classifier detects but MSP reads as normal. On LOACO D2, MSP is below 0.5 for every class on both backbones (0.163 AHSD, 0.162 GRU), while P(attack) reaches 0.999 on the same models: the held-out attacks are detected with high confidence, and MSP reads that confidence as normality. On LOACO D3 MSP is again below 0.5 for every class; on natural novelty D3 it scores 0.262 and 0.245 against 0.853 and 0.918. The inversion appears in-distribution too, where MSP ranges from 0.075 to 0.394 (AHSD) and 0.065 to 0.340 (GRU) on grouped D1–D3 splits while P(attack) is 0.812–0.999 and 0.839–1.000. MSP exceeds P(attack) in only 5 of 34 evaluation–backbone pairs, all of them rolling cuts where P(attack) itself is at or near chance (0.345–0.548), i.e., where the classifier misses the novel class. On purged D5 the GRU's MSP is not inverted (0.889) but still below its P(attack) (0.999). **Answer to RQ1:** on binary detectors, MSP and energy measure certainty, not attack likelihood; the attack probability or a representation distance should be used instead.

B. C2: Rankings depend on protocol, dataset and cut-point

The leading novelty family changes across evaluations (Fig. 3, framed cells): representation-distance scores lead five of the seven fixed evaluations and benign-only detectors two, and the second backbone selects the same leader in all seven. Rolling cut-points show how fragile any single choice is (Fig. 3, right). Pooled over 45 units, benign-only detectors lead (0.748 [0.699, 0.796]) ahead of representation scores (0.623 [0.566, 0.683]; difference +0.125 [+0.048, +0.204]); against the GRU representation family the lead is smaller (+0.046 [−0.018, +0.118]). Per dataset the leader is representation on D1 and D5 but benign-only on D3, and per cut fraction it moves between representation, benign-only and CLAD. The supervised reference is strong on fixed evaluations, where XGBoost matches or beats the best novelty family in four of seven (e.g., LOACO D2 0.998, natural novelty D3 0.923), but fragile across cuts: pooled it trails the leading family by −0.076 [−0.154, −0.009], and on D3 at f=0.45 it scores 0.263, below chance. CLAD never leads a fixed evaluation and trails benign-only detectors over rolling units (−0.096 [−0.167, −0.030]); our reimplementation reaches 0.972 on purged D5, close to the 0.997 its authors report on the same dataset [22]. **Answer to RQ2:** a family chosen on one protocol, dataset or cut can be the wrong choice on another; no family, and not a supervised classifier, is safe.

C. C3: Natural novelty exposes hidden instability

Representation-distance scores lose most stability under natural novelty (Fig. 4 shows every seed on ToN-IoT): their mean seed standard deviation rises 6.1–17.7× on the first backbone and 2.7–2.8× on the second. Confidence scores change less (1.4–2.1× and 1.1–1.2×). The effect is concentrated on D3: over rolling units the seed standard deviation of kNN is 0.185

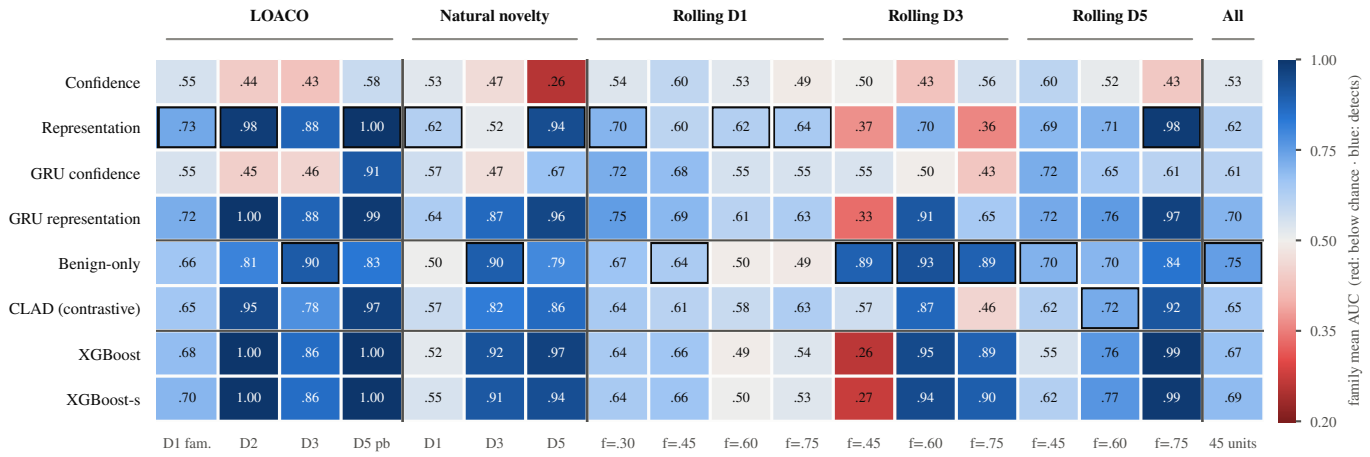


Fig. 3. Family mean AUC for every evaluation (columns) and detector family (rows; GRU families from the second backbone). Colour is centred at chance: red cells are below 0.5 (the family ranks attacks as *more* normal than benign traffic), blue cells detect. A black frame marks the leading novelty family of the first backbone in each column; it moves between protocols, datasets and cut-points.

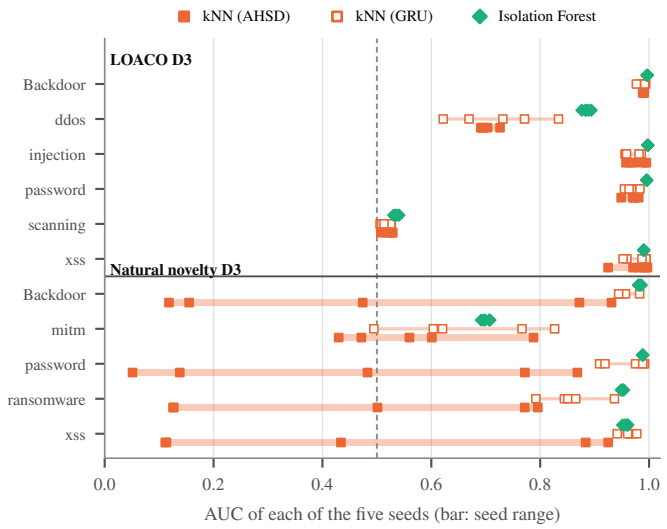


Fig. 4. Every seed drawn: AUC of each of the five training seeds per ToN-IoT class. Under LOACO (top) the seeds of a detector agree; under natural novelty (bottom) the same kNN detector ranges from below chance to near-perfect, while the seeded Isolation Forest stays put.

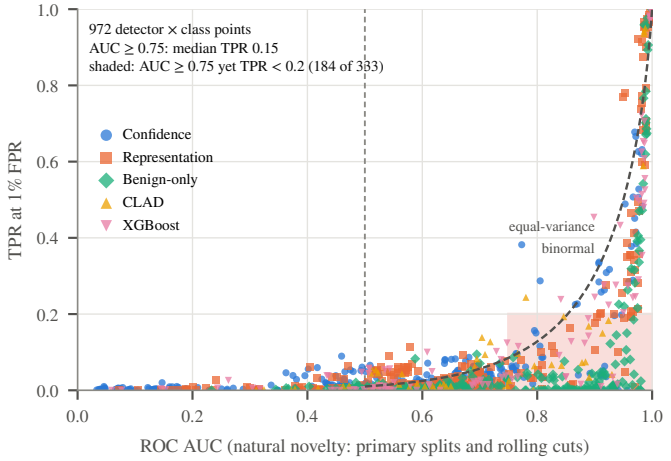


Fig. 5. AUC vs detection at 1% FPR for every detector and natural-novelty class (primary splits and rolling cuts; mean over seeds). Dashed: what an equal-variance binormal score with that AUC would detect. Shaded: AUC $\geq$ 0.75 yet fewer than 20% of attack windows caught.

on D3 against 0.036 on D1 and 0.032 on D5, and 0.154 for the GRU's kNN. Seeded benign-only detectors stay below 0.02 (Isolation Forest 0.008, autoencoder 0.017), and XGBoost-s at 0.014, so the instability belongs to learned representations under shift, not to seeding as such. **Answer to RQ3:** the zero-day result of a representation score can depend on the training run as much as on the detector; LOACO hides this.

D. C4: AUC overstates zero-day coverage

AUC and detection at an operating point disagree sharply (Fig. 5). Of the 333 detector–class points with an AUC of at least 0.75, 184 catch fewer than 20% of attack windows at 1% FPR (median 0.15), and most fall below the binormal curve: novel attacks overlap the benign score tail. Under LOACO D2 the best families detect 87–99% of held-out windows at 1% FPR. Under natural novelty D3, benign-only detectors reach an AUC of 0.90 but detect only 12% at 1% FPR, and XGBoost 38%. Over the 45 rolling units no family exceeds 18% at 1% FPR (XGBoost 0.18; benign-only 0.09 at AUC 0.75), and PR-AUC stays at or below 0.41. **Answer to RQ4:** a zero-day AUC of 0.7–0.9 can mean that most of a new attack goes unnoticed at a false-alarm rate an operator can afford.

E. Ablations

Table IV varies one factor at a time. The *backbone* does not change which family leads, but it changes absolute values: on natural novelty D3 the first backbone's representation scores fall to chance (0.522) while the GRU's reach 0.869, still below benign-only detectors (0.898). The *checkpoint* moves representation and CLAD family means by at most 0.073 and

TABLE IV
ABLATIONS: ONE FACTOR VARIED AT A TIME, ALL ELSE FIXED (FAMILY MEAN AUC; 95% BOOTSTRAP INTERVALS).

Factor	Effect
Backbone (AHSD→GRU)	Same leading family in 7/7 evaluations; Repr.−GRU-r NN D3 −0.347 [−0.504, −0.163], rolling −0.079 [−0.121, −0.038]
Checkpoint (best→final)	max $\|\Delta\|$ family AUC: Repr./CLAD 0.073, Conf. 0.205; GRU-r 0.022, GRU-c 0.189
XGBoost seeding (fixed→s)	$\geq$ leader in 4 vs 3 of 7; rolling AUC 0.672 vs 0.686; seed std 0 vs 0.019
Split (grp.→temp.)	P(attack) D3 AHSD 0.975→0.664, GRU 0.981→0.857, XGB 0.977→0.978
Cut fraction f	Leading family per cut: 0.30: Repr., 0.45: Benign, 0.60: CLAD, 0.75: Benign
Dataset (rolling)	Leading family: D1: Repr., D3: Benign, D5: Repr.; Benign−Repr. pooled +0.125 [+0.048, +0.204]

confidence scores by up to 0.205 (D5, whose validation set holds 26 attack windows). *Seeding XGBoost* barely changes its mean, so the stability comparison of C3 is not an artifact of a deterministic baseline. The *temporal split* degrades the first backbone on D3 (0.975→0.664) much more than the GRU (0.981→0.857), while XGBoost holds. The *cut fraction* and the *dataset* change the leading family, as described in C2.

F. Comparison with previously reported results

Table V sets our numbers beside published ones on the same datasets or questions; metrics and dataset versions differ, so it shows direction rather than a ranking. Class hold-out results in the literature (0.97–0.997 AUROC; 87–92% zero-day detection) are matched by our LOACO numbers on D2 and D5, but the same detectors drop under natural novelty and rolling cuts: CLAD from 0.972 (purged D5) to 0.702 (rolling D5), and XGBoost on ToN-IoT from 0.862 (LOACO) to 0.699 (rolling). Softmax entropy and energy were reported at 0.76–0.80 AUROC on multi-class NF-ToN-IoT [14]; on binary detectors the same scores invert (0.26–0.27), as C1 predicts. Temporal degradation [15], [16] and near-chance transfer [30] are confirmed, and Zoppi et al.'s low unknown-attack recall for XGBoost (0.38) [21] is matched by its 18% TPR at 1% FPR on rolling cuts.

G. Confirmatory results: temporal and cross-dataset shift

Temporal-split effects are dataset-specific (Fig. 6, left): on D1 both the first backbone and XGBoost lose 0.27–0.29 AUC, on D2 neither changes by more than 0.014, and on D5 both improve. Zero-shot transfer collapses towards chance, as reported before [29], [30]: the median of 108 transfer results (12 detectors × 9 pairs) is 0.500, with 67% in [0.3, 0.6]; the highest come from D1→D3 (CLAD 0.876, XGBoost 0.832).

H. C5: Pre-registered hypotheses that failed

All four confirmatory hypotheses failed against their frozen gates (Table VI): a spectral model of the recurrent filter did not predict per-class detectability (P1); an adaptive equilibrium did not absorb persistent attacks (G1); benign drift with re-anchoring (G2) and benign-only superiority restricted to natural novelty (G3) also failed. G3's supervised family included

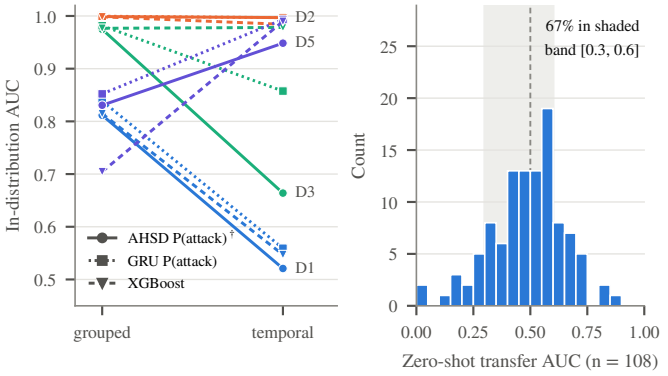


Fig. 6. Left: in-distribution AUC under grouped vs temporal splits (solid: AHSD P(attack); dotted: GRU P(attack); dashed: XGBoost). Right: the 108 zero-shot transfer AUCs cluster around chance.

MSP and energy, which C1 shows are misaligned, so G3 tested a flawed contrast; its verdict stands as recorded.

VII. DISCUSSION

Why the protocols disagree. LOACO removes one class but keeps the rest of the training distribution, including benign traffic from the hours of the test set. Natural novelty changes the attack class, the benign traffic and often the hosts at once. Supervised representations are shaped by the attacks they were trained to separate: under small shift they place a held-out class far from benign traffic (representation 0.977 on D2 LOACO), but when benign traffic drifts, as on D3, they become backbone- and seed-dependent. Benign-only detectors do not depend on which attacks were seen, which explains their stability and their lead over rolling cuts, but not a general superiority: they trail by 0.17 on D2 LOACO. XGBoost's strength on fixed evaluations agrees with evidence that tree ensembles are hard to beat on tabular data [44]; its collapse at some cuts shows that a supervised boundary can also exclude a new class on the wrong side.

Implications for security management. For an operator, a NIDS is a service whose assurance depends on evidence gathered before deployment. The usual evidence, one LOACO run scored by AUC with MSP or energy, can select the wrong family, raise alarms on the wrong windows, hide seed dependence and overstate coverage at an affordable alarm rate. We recommend the following checklist:

1) Never use max-softmax or energy as novelty alarms on binary detectors; use the attack probability or a representation distance.
2) Evaluate under natural temporal novelty at several cut-points wherever timestamps allow, alongside LOACO, and report when LOACO is infeasible.
3) Report TPR at the false-positive rate the security team can absorb, not only AUC.
4) Train at least five seeds and report per-class seed variability; reject detectors whose zero-day coverage depends on the seed.

TABLE V

PREVIOUSLY REPORTED RESULTS VS OURS ON THE SAME DATASETS OR QUESTIONS. METRICS, SPLITS AND DATASET VERSIONS DIFFER (STATED IN EACH CELL), SO THE COMPARISON SHOWS DIRECTION, NOT A LIKE-FOR-LIKE RANKING.

Setting	Source	Reported	Ours (this benchmark)
Zero-day hold-out, UNSW-NB15	[20]	Z-DR 87–92% (NF-v2)	LOACO D2: XGB 0.998, Repr. 0.977 AUC
CLAD, LYCOS-IDS2017	[22]	AUROC 0.997	purged 0.972; NN 0.864; rolling 0.702
Held-out families, ToN-IoT	[18]	AUROC 0.971	best: LOACO 0.903; rolling 0.903
Softmax entropy/energy, NF-ToN-IoT	[14]	AUROC 0.76 / 0.80	LOACO D3 MSP 0.269, energy 0.258
Random→temporal, TON-IoT	[15]	macro-F1 0.87→0.58	AUC: XGB 0.977→0.978, AHSD 0.975→0.664
Chronological, CIC-IDS	[16]	macro-F1 ≈0.50	temporal D1 AUC: XGB 0.547
Cross-dataset	[30]	AUROC 0.59–0.82	median 0.500 over 108 pairs×detectors
Unknown-attack recall	[21]	XGB Rec-Unk 0.38	rolling XGB TPR@1% FPR 0.18

TABLE VI

PRE-REGISTERED CONFIRMATORY GATES, EVALUATED AS FROZEN. ALL FOUR RETURNED STOP.

Gate	Hypothesis	Frozen rule	Observed
P1	Spectral model predicts per-class detectability	$\rho \geq 0.6$	$\rho = 0.326$
G1	Adaptive equilibrium absorbs persistent attacks	$\rho(\gamma, d') \leq -0.8$	-0.14 / -0.14 / 0.00
G2	Benign drift inverts scores; re-anchoring fixes it	≥5/9 detectors	4, 5, 0 of 9
G3	Benign-only beat supervised only under NN	$B - S \geq 0.15$ / ≤ 0.05	+0.191 / +0.269

5) Include a benign-only detector and a gradient-boosted tree as baselines.
6) Group flows into time blocks before windowing, deduplicate windows, purge neighbouring blocks on short captures, and pre-register acceptance criteria.

Limitations and threats to validity. Backbones: both backbones are recurrent; attention-based models may behave differently. *Dependence:* rolling units of one class at several cuts share test windows, so the 45 units are not independent; we also report per-dataset and per-unique-class results in the release. *Thin validation on D5:* 26 attack windows make checkpoint selection fragile for confidence scores. *Design choices:* the 1-hour grouping, any-attack window label, thresholds and 5:1 test ratio were fixed in advance but are not the only reasonable choices. *Post-hoc addition:* P(attack) is labelled and is not confirmatory evidence. *Reimplementation:* CLAD may differ from its authors' code, although it reproduces their D5 level. *Testbed data:* natural novelty follows each capture's schedule, which is closer to deployment than LOACO but is not live traffic.

VIII. CONCLUSION

Evaluating learned NIDS on natural zero-day traffic, rather than on one class hold-out run scored by AUC, changes what an operator should believe. On two backbones, max-softmax and energy inverted because confident detection reads as normality; detector-family rankings moved with protocol, dataset and cut-point, and a supervised XGBoost that matched the best novelty family on fixed evaluations trailed benign-only detectors over rolling cuts; representation scores became seed-dependent under natural novelty; and AUCs up to 0.75 meant at most 18% detection at 1% false-positive rate. Four pre-registered hypotheses failed and are reported in full. Code, splits and amendments are at [artifact URL].

REFERENCES

[1] R. Sommer and V. Paxson, "Outside the closed world: On using machine learning for network intrusion detection," in *Proc. IEEE S&P*, 2010, pp. 305–316.

[2] D. Arp *et al.*, "Dos and don'ts of machine learning in computer security," in *Proc. USENIX Security*, 2022, pp. 3971–3988.

[3] F. Pendlebury, F. Pierazzi, R. Jordaney, J. Kinder, and L. Cavallaro, "TESSERACT: Eliminating experimental bias in malware classification across space and time," in *Proc. USENIX Security*, 2019, pp. 729–746.

[4] G. Apruzzese, P. Laskov, and J. Schneider, "SoK: Pragmatic assessment of machine learning for network intrusion detection," in *Proc. IEEE EuroS&P*, 2023, pp. 592–614.

[5] M. Husák, D. Manoj, and P. Kumar, "Machine learning in intrusion detection: An operational perspective," in *Proc. CNSM*, 2024, pp. 1–7.

[6] G. Engelen, V. Rimmer, and W. Joosen, "Troubleshooting an intrusion detection dataset: The CICIDS2017 case study," in *Proc. IEEE SPW*, 2021, pp. 7–12.

[7] L. Liu, G. Engelen, T. Lynar, D. Essam, and W. Joosen, "Error prevalence in NIDS datasets: A case study on CIC-IDS-2017 and CSE-CIC-IDS-2018," in *Proc. IEEE CNS*, 2022, pp. 254–262.

[8] A. Rosay, F. Carlier, E. Cheval, and P. Leroux, "From CIC-IDS2017 to LYCOS-IDS2017: A corrected dataset for better performance," in *Proc. IEEE/WIC/ACM WI-IAT*, 2021, pp. 570–575.

[9] R. Flood, G. Engelen, D. Aspinall, and L. Desmet, "Bad design smells in benchmark NIDS datasets," in *Proc. IEEE EuroS&P*, 2024, pp. 658–675.

[10] L. D'hooge, M. Verkerken, B. Volckaert, T. Wauters, and F. De Turck, "Establishing the contaminating effect of metadata feature inclusion in machine-learned network intrusion detection models," in *Proc. DIMVA*, LNCS, Springer, 2022, pp. 23–41.

[11] U. Sabeel, S. S. Heydari, K. El-Khatib, and K. Elgazzar, "Unknown, atypical and polymorphic network intrusion detection: A systematic survey," *IEEE Trans. Netw. Serv. Manag.*, vol. 21, no. 1, pp. 1190–1212, 2024.

[12] G. Baye, P. Silva, A. Broggi, L. Fiondella, N. D. Bastian, and G. Kul, "Performance analysis of deep-learning based open set recognition algorithms for network intrusion detection systems," in *Proc. IEEE/IFIP NOMS*, 2023, pp. 1–6.

[13] Z. Zhang, Y. Zhang, D. Guo, and M. Song, "A scalable network intrusion detection system towards detecting, discovering, and learning unknown attacks," *Int. J. Mach. Learn. Cybern.*, vol. 12, no. 6, pp. 1649–1665, 2021.

[14] J. Talpini, F. Sartori, and M. Savi, "Enhancing trustworthiness in ML-based network intrusion detection with uncertainty quantification," *J. Reliable Intell. Environ.*, vol. 10, no. 4, pp. 501–520, 2024.

[15] M. Abdessamad and E. B. Tazi, "Feature selection versus feature extraction for IoT intrusion detection under temporal evaluation," *Int. J. Adv. Comput. Sci. Appl.*, vol. 17, no. 8, 2026.

[16] Z. Moczkodan and H. Ragab, "Do transformers actually help intrusion detection? A temporal sequence evaluation on CIC-IDS2017," arXiv:2606.11098, 2026.

[17] S. Mesca, E. Pricop, and G. Stamatescu, "Rigorous evaluation of machine learning intrusion detection for water treatment systems on SWaT network traffic," *Appl. Sci.*, vol. 16, no. 14, Art. 7206, 2026.

[18] S. Nayak, "ThreatFormer-IDS: Robust transformer intrusion detection with zero-day generalization and explainable attribution," arXiv:2603.00185, 2026.

[19] A. Alhalangy and G. E. A. Eltayeb, "Drift-aware continual generative behavioral detection for zero-day cyberattacks: Reproducible temporal evaluation and SIEM-ready alerting," *Information*, vol. 17, no. 9, Art. 852, 2026.

[20] M. Sarhan, S. Layeghy, M. Gallagher, and M. Portmann, "From zero-shot machine learning to zero-day attack detection," *Int. J. Inf. Secur.*, vol. 22, no. 4, pp. 947–959, 2023.

[21] T. Zoppi, A. Ceccarelli, T. Puccetti, and A. Bondavalli, "Which algorithm can detect unknown attacks? Comparison of supervised, unsupervised and meta-learning algorithms for intrusion detection," *Comput. Secur.*, vol. 127, Art. 103107, 2023.

[22] J. Wilkie, H. Hindy, C. Michie, C. Tachtatzis, J. Irvine, and R. Atkinson, "A novel contrastive loss for zero-day network intrusion detection," *IEEE Trans. Netw. Serv. Manag.*, vol. 23, pp. 2064–2076, 2026.

[23] M. Jodayree, A. Kavoosi Ghafi, S. Amiri, and P. Shaykholeslami, "Explainable zero-day attack detection in IoMT using transformer-based time-series modeling," *Sci. Rep.*, vol. 16, Art. 23252, 2026.

[24] M.-T. Thi and M. Gurusamy, "Multi-dimensional cross-granularity open-set network intrusion detection," *IEEE Trans. Netw. Serv. Manag.*, vol. 23, pp. 4632–4646, 2026.

[25] G. Andresini, F. Pendlebury, F. Pierazzi, C. Loglisci, A. Appice, and L. Cavallaro, "INSOMNIA: Towards concept-drift robustness in network intrusion detection," in *Proc. ACM AISec*, 2021, pp. 111–122.

[26] S. Yang, X. Zheng, J. Li, J. Xu, X. Wang, and E. C. H. Ngai, "ReCDA: Concept drift adaptation with representation enhancement for network intrusion detection," in *Proc. ACM KDD*, 2024, pp. 3818–3828.

[27] X. Zhang *et al.*, "Continual learning with strategic selection and forgetting for network intrusion detection," in *Proc. IEEE INFOCOM*, 2025, pp. 1–10.

[28] W. W. Lo, S. Layeghy, M. Sarhan, M. Gallagher, and M. Portmann, "E-GraphSAGE: A graph neural network based intrusion detection system for IoT," in *Proc. IEEE/IFIP NOMS*, 2022, pp. 1–9.

[29] S. Layeghy and M. Portmann, "Explainable cross-domain evaluation of ML-based network intrusion detection systems," *Comput. Electr. Eng.*, vol. 108, Art. 108692, 2023.

[30] M. Cantone, C. Marrocco, and A. Bria, "Machine learning in network intrusion detection: A cross-dataset generalization study," *IEEE Access*, vol. 12, pp. 144489–144508, 2024.

[31] M. Luay, S. Layeghy, S. Hosseininoorbin, M. Sarhan, N. Moustafa, and M. Portmann, "Temporal analysis of NetFlow datasets for network intrusion detection systems," arXiv:2503.04404, 2025.

[32] M. Sarhan, S. Layeghy, N. Moustafa, and M. Portmann, "NetFlow datasets for machine learning-based network intrusion detection systems," in *Proc. BDTA 2020*, LNICST 371, Springer, 2021, pp. 117–135.

[33] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization," in *Proc. ICISSP*, 2018, pp. 108–116.

[34] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems," in *Proc. MilCIS*, 2015, pp. 1–6.

[35] N. Moustafa, "A new distributed architecture for evaluating AI-based security systems at the edge: Network TON_IoT datasets," *Sustain. Cities Soc.*, vol. 72, Art. 102994, 2021.

[36] N. Koroniotis, N. Moustafa, E. Sitnikova, and B. Turnbull, "Towards the development of realistic botnet dataset in the Internet of Things for network forensic analytics: Bot-IoT dataset," *Future Gener. Comput. Syst.*, vol. 100, pp. 779–796, 2019.

[37] D. Hendrycks and K. Gimpel, "A baseline for detecting misclassified and out-of-distribution examples in neural networks," in *Proc. ICLR*, 2017.

[38] W. Liu, X. Wang, J. Owens, and Y. Li, "Energy-based out-of-distribution detection," in *Proc. NeurIPS*, 2020, pp. 21464–21475.

[39] K. Lee, K. Lee, H. Lee, and J. Shin, "A simple unified framework for detecting out-of-distribution samples and adversarial attacks," in *Proc. NeurIPS*, 2018.

[40] Y. Sun, Y. Ming, X. Zhu, and Y. Li, "Out-of-distribution detection with deep nearest neighbors," in *Proc. ICML*, 2022, pp. 20827–20840.

[41] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proc. IEEE ICDM*, 2008, pp. 413–422.

[42] B. Schölkopf, J. C. Platt, J. Shawe-Taylor, A. J. Smola, and R. C. Williamson, "Estimating the support of a high-dimensional distribution," *Neural Comput.*, vol. 13, no. 7, pp. 1443–1471, 2001.

[43] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. ACM KDD*, 2016, pp. 785–794.

[44] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why do tree-based models still outperform deep learning on typical tabular data?" in *Proc. NeurIPS Datasets and Benchmarks*, 2022, pp. 507–520.

[45] M. López de Prado, *Advances in Financial Machine Learning*. Hoboken, NJ, USA: Wiley, 2018, ch. 7.

[46] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor, "The preregistration revolution," *Proc. Natl. Acad. Sci. USA*, vol. 115, no. 11, pp. 2600–2606, 2018.